

AI-BFSI Internship Project Documentation

Demonstration of Proposed Features – Subscription Waste Detector

1. Introduction

Demonstrating proposed features validates that the functionality planned during requirement analysis and project proposal has been successfully implemented. It establishes a clear relationship between the proposed solution and the final application.

For the **Subscription Waste Detector**, the proposed features focused on improving financial awareness by detecting recurring subscription expenses, visualizing spending, providing renewal reminders, and generating AI-powered recommendations. The completed Version 1 implementation incorporates these capabilities using **Streamlit, Python, Pandas, Plotly, and the Groq API**.

2. Demonstration Objectives

- Verify that proposed functionality has been implemented
- Confirm each feature operates correctly
- Show how modules contribute to the overall solution
- Validate alignment between objectives and implementation
- Provide evidence of completed Version 1 functionality

3. Feature Demonstration Overview

Feature ID	Proposed Feature	Implementation Status
PF-01	CSV File Upload	Successfully Implemented
PF-02	Excel File Upload	Successfully Implemented
PF-03	Manual Expense Entry	Successfully Implemented
PF-04	Expense Data Processing	Successfully Implemented
PF-05	Subscription Detection	Successfully Implemented
PF-06	Subscription Listing	Successfully Implemented
PF-07	Monthly Spending Analysis	Successfully Implemented
PF-08	Interactive Dashboard	Successfully Implemented
PF-09	Plotly Visualizations	Successfully Implemented
PF-10	Renewal Reminder	Successfully Implemented
PF-11	AI-Powered Recommendations	Successfully Implemented

4. Feature Demonstrations

PF-01: CSV File Upload

Demonstration: Users upload CSV files via Streamlit; records are validated and processed.

Outcome: CSV expense records imported successfully.

PF-02: Excel File Upload

Demonstration: Excel (.xlsx) files uploaded and processed using OpenPyXL. **Outcome:**

Expense information prepared for subscription detection.

PF-03: Manual Expense Entry

Demonstration: Users manually enter expense details in the interface. **Outcome:** Manual records included in analysis.

PF-04: Expense Data Processing

Demonstration: Python + Pandas organize raw records into structured datasets. **Outcome:** Data prepared for recurring subscription analysis.

PF-05: Subscription Detection

Demonstration: Application identifies recurring payments based on transaction patterns.

Outcome: Recurring subscriptions detected and summarized.

PF-06: Subscription Listing

Demonstration: Subscriptions displayed with name, amount, category, and renewal date.

Outcome: Organized subscription summary available.

PF-07: Monthly Spending Analysis

Demonstration: Aggregates subscription costs to compute monthly expenditure. **Outcome:** Clear monthly spending summary provided.

PF-08: Interactive Dashboard

Demonstration: Streamlit dashboard displays summaries, reminders, and recommendations.

Outcome: Centralized interface for subscription management.

PF-09: Plotly Visualizations

Demonstration: Interactive charts show spending trends and categories. **Outcome:** Financial data visualized effectively.

PF-10: Renewal Reminder

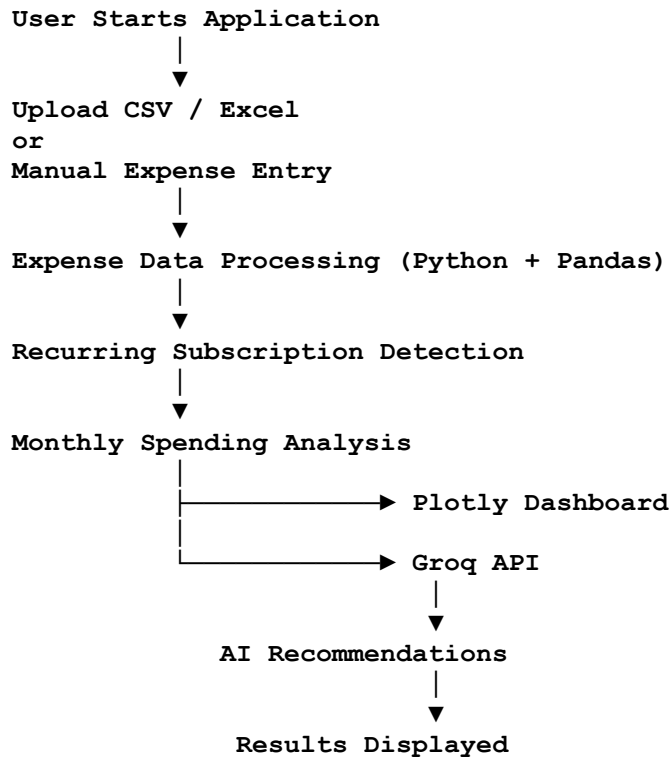
Demonstration: Application highlights subscriptions nearing renewal. **Outcome:** Users receive timely reminders.

PF-11: AI-Powered Recommendations

Demonstration: Subscription summaries sent to Groq API; recommendations returned.

Outcome: Personalized cost-saving suggestions displayed. **Note:** No custom AI model training; Groq API inference only.

5. End-to-End Feature Workflow



6. Feature Validation Summary

Feature	Validation Result
CSV Upload	Successfully Tested
Excel Upload	Successfully Tested
Manual Entry	Successfully Tested
Expense Processing	Successfully Tested
Subscription Detection	Successfully Tested
Dashboard	Successfully Tested
Plotly Charts	Successfully Tested
Renewal Reminder	Successfully Tested
AI Recommendations	Successfully Tested

7. Features Outside Current Scope

The following were proposed as future enhancements but are **not included** in Version 1:

- Authentication
- Database integration

- Cloud synchronization
- Bank API integration
- Email notifications
- Mobile application
- Multi-user support
- Predictive analytics

8. Conclusion

The demonstration confirms that all core features proposed during planning have been successfully implemented. The Subscription Waste Detector enables users to upload expense data, detect recurring subscriptions, analyze monthly spending, visualize financial information, monitor renewal dates, and receive AI-powered recommendations.